

Smit Merchant

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SUMMARY

Full-stack developer with a Master's in Artificial Intelligence from Sheffield Hallam University and a background in Computer Engineering. Experienced in designing and shipping production-ready applications that combine modern web technologies with practical AI integration. Demonstrated ability to take projects from concept to deployment across a range of domains including lead generation, food transparency, mental health, and academic research. Comfortable working across the full stack with hands-on experience in Next.js, Python, FastAPI, AWS Bedrock and Google Cloud, alongside AI integration

WORK EXPERIENCE

Adtecher <i>AI Full Stack Developer</i>	Sheffield, UK <i>Feb 2026 - Present</i>
- Built and shipped a five-phase AI-powered sales intelligence pipeline that automates prospect discovery, lead scoring, and personalised outreach generation at scale. - Worked across the full stack, Python and FastAPI backend through to a Next.js frontend dashboard with real-time pipeline monitoring. - Responsible for ongoing bug fixing and maintaining system stability across the product.	

EDUCATION

Sheffield Hallam University <i>Master's In Artificial Intelligence</i>	Sheffield, UK <i>Graduation Date: Jan 2025</i>
University Of Mumbai <i>Bachelor's in Computer Engineering</i>	Mumbai, India <i>Graduation Date: Jul 2023</i>

PROJECT EXPERIENCE

Independent Projects

LeadGen Pro

- Developed a web application that automates the **end-to-end lead generation and outreach workflow** for local businesses.
- Enabled users to **search businesses by location** and identify high-potential leads lacking an online presence (e.g., no website).
- Built an AI-powered feature to **generate full website prototypes instantly** for selected businesses.
- Integrated AI to create **personalised cold outreach emails** linking to generated websites.
- Implemented a **lead management system** with tracking statuses (e.g., Contacted, Replied) to streamline follow-ups.
- Focused on improving outreach efficiency by combining **automation, AI generation, and CRM-like tracking**.

Food Transparency – AI-Based Food Health Analysis

- Built a web application that analyses food products and scores them from 0–100 based on nutritional value and ingredient composition.
- Integrated DeepSeek API to break down complex ingredients into simple, user-friendly health insights.

ReflectoChain – AI-Powered Thought Tracking

- Developed an application that enables users to **log their thoughts periodically (e.g., every 6 hours)**, creating a structured “chain of thoughts.”
- Implemented AI-based summarisation to generate **weekly summaries of user thoughts without altering original entries**.
- Designed a **therapist dashboard** to present aggregated insights, helping professionals understand user patterns and mental state.
- Focused on maintaining **data integrity and ethical AI usage**, ensuring summaries remain faithful to original inputs.
- Explored the intersection of **AI, mental health, and behaviour tracking** for real-world applications.

ACADEMIC PROJECT EXPERIENCE

Sheffield Hallam University <i>Dissertation: Action Recognition for Surveillance</i>	Sheffield, UK <i>Sep 2024 - Jan 2025</i>
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- Designed a deep learning model (ResNet-34 + 3D CNN) to detect and classify human activities in real-time surveillance footage.
- Achieved **89.4% classification accuracy**, precision of **88.7%**, and **31 FPS** real-time performance.
- Explored **ethical AI** and **privacy-preserving techniques** such as federated learning.
- Suggested future improvements including multi-modal data integration and lightweight scalable models.

Sheffield Hallam University *by (Head of Computing)*

Sheffield, UK

Smart Search for Research Expert Connections

Jul 2025 - Sep 2025

- Served as Quality Assurance & AI Tester for an AI-driven platform connecting 100+ researchers across the university.
- Investigated PCA and alternative AI techniques to enhance usability, researcher clustering, and matchmaking accuracy.
- Conducted extensive testing and quality assurance, ensuring system reliability, accuracy, and smooth user experience.
- Collaborated with developers to evaluate NLP-based approaches for automated researcher profiling and recommendations.

SKILLS

Programming & Frameworks	Python, PyTorch, TensorFlow, scikit-learn,
MLOps & Deployment	FastAPI, Docker, GitHub Actions, REST API Development
Data & Tools	SQLNumPy, OpenCV
Core Expertise	NLP, Computer Vision, API Development
Other Skills	Problem-solving & analytical thinking, Communication & teamwork, Adaptability in cross cultural teams, Time management & project delivery under deadlines